Part A. PERSONAL INFORMATION

CVA date	11/11/2026

First name	Rodrigo		
Family name	Granjel Rodríguez		
Gender	Male	Birth date	09/07/1993
ID number	53817055E		
Email	rodrigo.granjel@bc3research.org	URL Web	https://granjel.github.io/
Open Researcher and Contributor ID (ORCID)		0000-0002-7785-4106	

A.1. Current position

Position	Postdoctoral researcher		
Start date	15/05/2023 (42 months*)		
Institution	BC3 – Basque Centre for Climate Change		
Department	Ecosystems and Global Change		
Country	Spain	Phone number	+34 667 04 86 44
Keywords	Community ecology; Species coexistence; Biotic interactions; Global change; Ecosystem stability and recovery		

*Including 5 months of childcare leave in separate periods between 10/2025 and 09/2026.

A.2. Previous positions

Period	Position / Institution / Country
06/2022 – 03/2023 (10 months)	Fulbright research fellow / North Carolina State University / USA
11/2017 – 03/2022 (53 months)	FPU doctoral researcher / Universidad de Sevilla / Spain
09/2017 – 10/2017 (2 months)	JAE-INTRO research assistant / IRNAS-CSIC / Spain

A.3. Education

Degree / Master's / PhD	University / Country	Year
PhD in Integrative Biology	Universidad de Sevilla / Spain	2023
MSc in Terrestrial Ecology	Universitat Autònoma de Barcelona / Spain	2017
BSc in Biology	Universidad de Vigo / Spain	2016

Part B. CV SUMMARY

I study how species interactions shape coexistence, ecosystem functioning and recovery under environmental change. I combine coexistence theory and ecological network approaches with experiments and quantitative models. My MSc research linked soil organisms and environmental conditions to carbon exports from peatlands (*Geoderma*, 2020).

During my **FPU-funded PhD**, I investigated how environmental change affects coexistence beyond pairs of species. With Eric Allan in Bern, I co-designed experiments, analysed data and wrote a paper showing that nutrient and pathogen effects depend on the number of interacting species (*New Phytologist*, 2023). During five months of visits supported by the University of Bern, I helped maintain the PaNDiv experiment. This collaboration continues with a manuscript on land-use intensification and coexistence, under review. In *Oikos* (2026), I led analyses, software development and writing to link simple plant interaction patterns to coexistence in a complex plant–herbivore community.

In parallel, I developed a chemical-ecology collaboration outside my doctoral group through visits to MBG-CSIC, including one funded by a competitive FPU mobility grant. I conceived and led the **AEET-funded CoVOCom project** to investigate how chemical communication affects plant responses to herbivory. Its experiments showed that volatile signalling reduces damage in *Arabidopsis* (*Environmental and Experimental Botany*, 2026). I am **first and corresponding author of these three papers**. Work at MBG also produced a *Plant, Cell & Environment* paper (2021) from an LMU Munich BSc thesis I co-supervised.

Further collaborations extended my research to aquatic and microbial communities. A five-month Santander-funded stay with Tadeu Siqueira at UNESP (Rio Claro, Brazil) developed into continuing research with UNESP and UFSCar on plankton coexistence under agrochemical stress; I am **last and corresponding author** of the manuscript in preparation. During a **ten-month Fulbright fellowship** with William Petry at North Carolina State University, I established protist experiments for a project

I conceived on microbial interactions, stability and functioning. Two manuscripts nearing submission test predictions of pairwise competition and multispecies coexistence against experimental data. The laboratory and Bayesian modelling skills I developed there now inform my postdoctoral research.

At BC3, with Daniel Montoya within ERC Consolidator RECODYN, I now ask **which routes of community recovery are possible under warming**. I develop Bayesian models to infer plant growth and interactions from experimental biomass data and, with Jürg Spaak at RPTU (Germany), connect these estimates to assembly graphs describing possible sequences of species establishment and community endpoints. I established this collaboration to link empirical recovery dynamics with coexistence theory, carrying uncertainty in interaction estimates into recovery predictions. Research visits to CNRS SETE (France; 173 days in 2023–2025) supported joint work with local teams using the Metatron facility. Through AgroServ's Horizon Europe access programme, ChEMISt connects this work with my chemical-ecology research using the same infrastructure. I am first author of two postdoctoral manuscripts in progress. I also contributed analyses, interpretation and writing to a study of functional diversity and stability across 235 grasslands (*Nature Communications*, 2026).

For **every published paper I have led**, I have archived well-documented data and reproducible analysis code in permanent public repositories, including Zenodo and Dryad. I have taught 192 hours at undergraduate and master's level, co-supervised an MSc thesis and mentored two DAAD RISE Worldwide interns at MBG-CSIC. I serve on the PCI Ecology recommenders panel, have evaluated Fulbright fellowships and AEET projects, and served on the scientific committee and coordinated a session at SIBECOL–AEET 2025. I organised my group's contribution to the European Researchers' Night, bringing together BC3 and EHU researchers to explain biodiversity and human pressures through games and visual activities for children and adults.

With Carlos Pérez Carmona, I aim to connect this experimental and modelling background with functional ecology to test how functional differences influence coexistence and recovery under environmental change.

Part C. RELEVANT MERITS

C.1. Publications

1. *Journal article:* **Granjel RR**, Gross N, Badenhausser I, Arroyo J, Godoy O. 2026. Simple plant–plant interaction patterns predict structural niche and fitness differences in a multitrophic grassland community. OIKOS, e12212. doi: 10.1002/oik.12212

   > Corresponding author and position 1/5 authors. I used statistical models to quantify key direct and higher-order interactions among plants and grasshoppers. We show that simple patterns of competition and facilitation between plants help explain coexistence within a complex food web.

2. *Journal article:* **Granjel RR**, Martín-Cacheda L, Röder G, Izquierdo-Ferreiro I, Martín-Díaz A, Picó FX. 2026. Plant–plant signalling reduces herbivore damage in two *Arabidopsis thaliana* populations with contrasting life histories. ENVIRONMENTAL AND EXPERIMENTAL BOTANY 251, 106472. doi: 10.1016/j.envexpbot.2026.106472

   > Corresponding author and position 1/6 authors. From my AEET-funded project. We show that volatile communication increases resistance to herbivory in two contrasting populations of *A. thaliana*, and suggest that herbivory may reshape links between chemical and life-history traits.

3. *Journal article:* Huang M, **Granjel RR**, Montoya D, Xiao Y, Liu X, Hautier Y. 2026. Plant diversity within communities, not among them, stabilizes grassland productivity across spatial scales. NATURE COMMUNICATIONS 17, 2145. doi: 10.1038/s41467-026-69028-5

   > Position 2/6 authors. I contributed to statistical analyses, interpretation and writing. We show that functional traits mediate the relationship between environmental conditions and ecosystem stability differently across spatial scales.

4. *Journal article:* Montoya D, Burón-Ugarte A, …, **Granjel RR**, …, Porras-Gómez J. 2025. Complexity and interpretability in global change ecology. PLOS CLIMATE 4, e0000587. doi: 10.1371/journal.pclm.0000587

   > Position 5/10 authors. First contribution at my BC3 postdoc. I helped conceptualise and write this piece on the complexity–interpretability trade-off in global change experiments.

5. *Journal article:* Jiménez-Mejías P, Manzano S, …, **Granjel RR**, …, Fitzpatrick M. 2024. Protecting stable biological nomenclatural systems enables universal communication: A collective international appeal. BIOSCIENCE 74, 467–472. doi: 10.1093/biosci/biae043
   > Position 1280/1563 authors. I endorsed this international appeal for stable biological nomenclature and universal scientific communication.
6. *Journal article:* **Granjel RR**, Allan E, Godoy O. 2023. Nitrogen enrichment and foliar fungal pathogens affect the mechanisms of multispecies plant coexistence. NEW PHYTOLOGIST 237, 2332–2346. doi: 10.1111/nph.18689
   > Corresponding author and position 1/3 authors. My first PhD paper, where I led all steps from data acquisition in Switzerland to manuscript preparation. We show that the effects of nutrients and pathogens on coexistence mechanisms need to be understood beyond pairwise interactions.
7. *Journal article:* Moreira X, **Granjel RR**, …, Rasmann S. 2021. Apparent inhibition of induced plant volatiles by a fungal pathogen prevents airborne communication between potato plants. PLANT, CELL & ENVIRONMENT 44, 1192–1201. doi: 10.1111/pce.13961
   > Position 2/10 authors. Derived from a BSc thesis I co-supervised at LMU Munich. We show that fungal infection determines how potato plants communicate via chemical volatiles.
8. *Journal article:* Juan-Ovejero R, **Granjel RR**, Ramil-Rego P, Briones MJI. 2020. The interplay between abiotic factors and below-ground biological interactions regulates carbon exports from peatlands. GEODERMA 368, 114313. doi: 10.1016/j.geoderma.2020.114313
   > Position 2/4 authors. Derived from my MSc thesis. We link abiotic conditions and interactions among plants, soil fauna and microbes to a clear peatland ecosystem function: carbon exports.
9. *Book chapter:* **Granjel RR**, Juan Ovejero R, Briones MJI. 2017. Diversidad edáfica en diferentes hábitats turbosos de la serra do Xistral. CONSERVACIÓN E XESTIÓN DE HUMIDAIS EN GALICIA. HORREUM-IBADER, 31–48. ISBN: 978-84-939661-4-0.
   > Position 1/3 authors. Derived from my BSc thesis. It represents the first characterisation of soil biodiversity across different peatland habitats in Serra do Xistral, Galicia.

C.2. Conferences

1. *Invited seminar:* **Granjel RR**. Starting over: Or the hard life of plants in a recovery dynamics experiment. Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau (Landau, Germany; June 2026).
2. *Scientific committee member, session coordinator and moderator:* The interplay between species coexistence and abiotic factors: community ecology (general session GSC). III SIBECOL & XVII AEET Meeting (Pontevedra, Spain; June 2025).
3. *Poster presentation:* **Granjel RR**, Montoya D. How does warming affect the coexistence mechanisms of recovering communities? III SIBECOL & XVII AEET meeting (Pontevedra, Spain; June 2025).
4. *Invited seminar:* **Granjel RR**. `Theory < data` for community ecology. Rheinland-Pfälzische Technische Universität Kaiserslautern-Landau (Landau, Germany; April 2025).
5. *Oral presentation:* **Granjel RR**, Godoy O. Environmental variability and the mechanisms behind species persistence. XV AEET National Conference (Plasencia, Spain; October 2021).
6. *Oral presentation:* **Granjel RR**, Allan E, Godoy O. The consequences of nitrogen enrichment and foliar pathogens for multispecies coexistence in grasslands. Festival of Ecology BES (Online due to the COVID-19 pandemic; December 2020).
7. *Poster presentation:* **Granjel RR**. Coexistence and VOC-mediated plant–plant communication in response to insect herbivory (CoVOCom). XVII ECOFLOR Meeting AEET (Bilbao, Spain; March 2020).
8. *Invited seminar:* **Granjel RR**. The struggle for coexistence: Studying biodiversity beyond pairwise interactions. Universität Bern (Bern, Switzerland; October 2019).
9. *Oral presentation:* **Granjel RR**, Gross N, Godoy O. How does the interaction network determine the herbivory effects on plant coexistence? 1st SIBECOL Meeting (Barcelona, Spain; February 2019).

10. *Poster presentation:* **Granjel RR**, Arroyo J, Godoy O. How do biotic interactions and environmental conditions shape species coexistence? Ecology Across Borders 2017 (Ghent, Belgium; December 2017).

C.3. Research projects

1. ChEMISt: Unveiling the factors shaping agroecosystem stability under global change: phytochemistry, functional traits, and multitrophic interactions. Ref. 24-C1-Agro-TNA-032. AgroServ Transnational Access, Horizon Europe. Co-PIs: Lucille T. S. Chrétien and Daniel Montoya, at BC3 (Spain); 2024–2026. Estimated value: €85,000. Awarded access to facilities and analytical services: CNRS-SETE (France) and CZU (Czech Republic). Ongoing.
   > I connect my research on coexistence and chemical ecology through experiments and fieldwork using the RECODYN infrastructure.
2. RECODYN: Ecosystem recovery dynamics and their response to climate change and habitat fragmentation. Ref. 101043548. ERC Consolidator Grant, European Research Council. PI: Daniel Montoya, at BC3 (Spain); 2023–2027. Total funding: €1,999,315. Ongoing.
   > I develop Bayesian models from experimental data to understand how plant interactions shape community recovery under warming.
3. SISMiC: The role of simple interactions in the stability and function of microbial communities. Predoctoral Research Fellowship, Fulbright Spain and Regional Government of Andalusia. PI: Rodrigo R. Granjel, at North Carolina State University (USA; William Petry's group); 2022–2023. Total funding: US$23,660. Completed.
   > I conceived this project to understand how simple assembly rules lead to complex, functional ecosystems, using experiments with microbial communities under lab conditions.
4. CoVOCom: Coexistence and chemical communication between plants in response to insect herbivory. Early-career Ecology Project Grants (“Tomando la iniciativa”), Spanish Association of Terrestrial Ecology (AEET). PI: Rodrigo R. Granjel, at Universidad de Sevilla and MBG-CSIC (Spain); 2020. Total funding: €2,000. Completed.
   > My first project as an independent PI. It combined my two main interests, interactions and coexistence, to understand how plants persist under herbivory via chemical communication.
5. MeDiNaS: Mechanisms of biodiversity maintenance in variable environments. Ref. RTI2018-098888-A-I00. Plan Nacional de I+D+i / Retos Investigación, Spanish Ministry of Science, Innovation and Universities. PI: Oscar Godoy, at Universidad de Cádiz (Spain); 2019–2021. Total funding: €105,270. Completed.
   > Led by my PhD supervisor. I contributed to fieldwork and discussions as a work-team member.

C.4. Knowledge and technology transfer

For every published paper I have led, I have openly archived well-documented data and reproducible analysis code in permanent repositories, including Zenodo and Dryad, enabling others to reproduce the analyses and reuse the materials.

In 2019, I provided a biological expert assessment for judicial proceedings before the court in Padrón (Galicia, Spain), addressing environmental and potential public health concerns associated with urban wildlife in a residential building.